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To cite this article: Ali Ekhlesi *et al* 2021 *Biomed. Phys. Eng. Express* **7** 025008

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PAPER

EEG signal analysis during Ishihara's test in subjects with normal vision and color vision deficiency

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Keywords: EEG signal processing, color vision deficiency (CVD), ishihara test, classification, frequency features

Supplementary material for this article is available [online](#)

RECEIVED
23 September 2020

REVISED
22 December 2020

ACCEPTED FOR PUBLICATION
14 January 2021

PUBLISHED
29 January 2021

Abstract

Color Vision Deficiency (CVD) is one of the most common types of vision deficiency. People with CVD have difficulty seeing color spectra depending on what types of retina photoreceptors are impaired. In this paper, the Ishihara test with 38 plates was used to examine the Electroencephalogram (EEG) of ten subjects with CVD plus ten healthy individuals. The recording was performed according to the 10–20 international system. The C-based software was programmed so that subjects could select the number or path in each test plate in the software options while recording EEG. Frequency features in different frequency bands were extracted from the EEG signals of the two groups during the Ishihara test. Statistically significant differences ($P < 0.05$) between features were assessed by independent samples t-test with False Discovery Rate (FDR) correction. Also, the K-nearest neighbor classifier (KNN) was used to classify the two groups. The results revealed that the most significant difference between the two groups in the Ishihara test images occurred for the electrodes located in the right temporoparietal areas (P4 and T6) of the brain in the Delta, Theta, Beta1, and Beta2 frequency bands. The KNN classifier, using the signals that reported the greatest statistical difference between the two groups, showed that the two groups were distinguishable with 85.2% accuracy. In this way, images from the Ishihara test that would provide the most accurate classification were identified. In conclusion, this research provided new insights into EEG signals of subjects with CVD and healthy subjects based on the Ishihara color vision test.

1. Introduction

Color Vision Deficiency (CVD) is one of the most important visual disorders affecting a considerable percentage of the population. A person with CVD fails to see a certain color under normal lighting conditions [1]. Color blindness is the most common vision disorder [2], with approximately, 8%–9.3% of men and 0.5%–3.2% of women suffering from this disorder [2–4]. This relatively significant percentage, especially in men, justifies the extensive research on the diagnosis and treatment of color blindness.

Color detection is a complex activity that depends not only on the eye and its components but also on the function of the visual cortex of the brain. The most basic stages of color processing occur in the retina. This region consists of two types of photoreceptor cells

named after their shape: rod and cone receptors. The first category enables us to see objects under dark and dim conditions, while the cone receptors perform color detection processes and operate at higher light levels. The sensitivity of the cone receptors to different wavelengths is variable, and are accordingly are divided into three types: S-cone (about 420 nm), M- Cone (about 530 nm), and L-cone (about 560 nm), which are more sensitive to blue, green and red respectively [5, 6].

The outcome of the combination of responses of all three cones enables us to see a specific color. Visual information is transferred from the retina to the primary visual cortex by the Lateral Geniculate Nucleus (LGN) in the thalamus. LGN serves as a relay station between the retina and the primary visual cortex located in the dorsal temporal lobe. Various studies have

already been conducted on LGN and how it affects the visual signal [5].

Recent studies have shown that the color center is neither isolated nor traceable to a single area in the visual cortex. In other words, color processing takes place in multiple areas that have different roles [5, 7, 8]. Several functional Magnetic Resonance Imaging (fMRI) studies have demonstrated that the color center begins in V1. Then for further processing, the signals are transferred to extrastriate areas V2 and V4 [9].

One or two types of retinal cones in people with color blindness may be impaired due to various factors such as genetic disorder, neurological disease, or other factors [10–12]. There are two types of color blindness:

- Monochromacy (total color blindness), a rare type of disorder in which the retina contains only one type of cone cell.
- Dichromacy, which refers to the condition where the function of one type of the cone receptors is impaired. The most common disorders include protanomaly and deuteranomaly where the effective red (L) receptors or green (M) receptors are defective and consequently, the person sees the red and green light color as dark or dark green [13, 14].

The Ishihara test is a very popular color perception test for red-green color deficiencies because of its ease of use and high accuracy in distinguishing between those with normal color vision and others with color vision defects [13, 15–20]. In this test, the colored dots are aligned such that CVD people cannot recognize the image as a picture with colored dots [15, 16]. Table 1 indicates four examples of Ishihara test images, along with a probable response for normal and CVD patients.

There are limitations to performed Ishihara tests through the display. Different quality of the display screens, a wide and varied range of colors and brightness levels can affect the results. In addition to the Ishihara test, there are other tests for color blindness that are less commonly used than the Ishihara test due to the difficulty of the test protocol [16, 17]. The results of the Ishihara test and other methods can be used to help CVD people achieve a better color perception, where color changes can enable them to differentiate between colors [21–24]. Today, these people can also benefit from equipment such as lenses and special glasses, and experience even more convenience [18, 25, 26].

In addition to disorders in the retina, brain function analysis can be used to assess the brain signal of people with color blindness. Electroencephalography (EEG) is an electrophysiological monitoring method to record the electrical activity of the brain. EEG measures voltage fluctuations resulting from ionic current within the neurons of the brain using surface

electrodes on the scalp [27, 28]. EEG is most often used to diagnose epilepsy, sleep disorders, depth of anesthesia, lie detection, brain death, and many aspects of brain functions. Compared to other brain imaging methods such as fMRI, EEG has a better temporal resolution.

Marey [16] presented a computer-based Ishihara test software on a Liquid Crystal Display (LCD) and achieved 100% sensitivity and 98.78% specificity. It has many benefits such as reducing the cost and time of the test and the possibility of EEG analysis during the test. Zhang and Zheng [29] analyzed the EEG signals for color stimuli and proposed a method of how to determine what color the subject get. Machado and Oliveria [30] proposed a model to consistently handle normal color vision, anomalous trichromacy, and dichromacy in a unified way by cones electrical response. Note, the studies that analyze color blindness utilizing the EEG signals are rare. Besides, none of these studies explain the differences in EEG signals of healthy and CVD patients. They have not specified either which image of the Ishihara test could better distinguish the two groups.

To the best of our knowledge, this is the first investigation to examine the differences in EEG between CVD subjects and healthy individuals during the Ishihara test. In this study, it was hypothesized that in addition to dysfunction of retinal receptors in CVD subjects, the two groups can also have significant differences in the EEG pattern. Consequently, frequency features have been investigated employing statistical tests to provide feature vectors for classification. In the 'Method' section, the procedure of data collection is defined. Then, the signal clearance and processing steps are presented. The 'Results' section deals with tables containing responses of the two groups of healthy and CVD people, along with the results of the statistical test. This section explains the results based on the electrodes and Ishihara images separately. Finally, the 'Discussion' Section concludes the results appropriately.

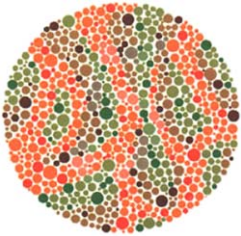
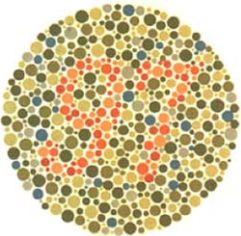
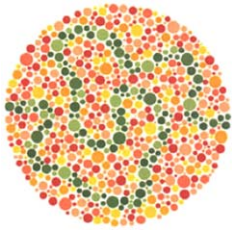
2. Method

In this section, firstly, the Ishihara test implementation is described. Afterward, the process of data collection is presented and finally, the processing and classification steps are explained.

2.1. Design of ishahara test and data

A 38-plate Ishihara test version was used. These plates involve both number recognition and path identification plates. The total sample size was 20 individuals (10 individuals in healthy (control) and 10 individuals in CVD (target) groups). All subjects were male and aged 26 ± 3.7 years. The subjects were collected by an experienced ophthalmologist based on a face-to-face interview, their medical profiles, and color vision

Table 1. Four samples of Ishihara test plates.

Image				
Normal	Nothing	97	35	Blue or green line
CVD	5	See something wrong or nothing	5 or 3	Nothing

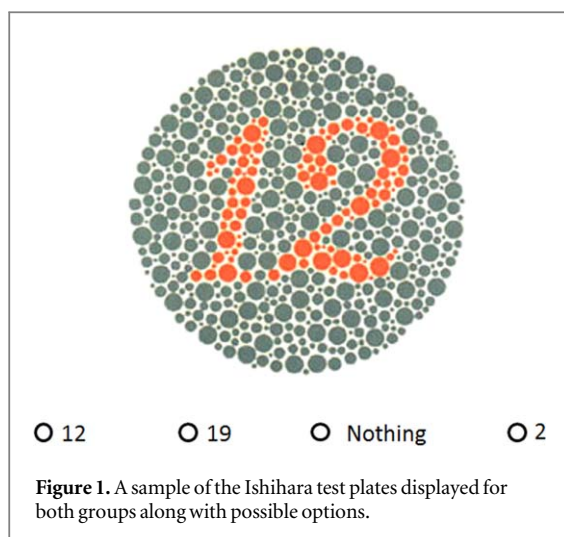


Figure 1. A sample of the Ishihara test plates displayed for both groups along with possible options.

evaluation. Accordingly, the green color deficit (deuteranopia) in the target group was confirmed by an ophthalmologist.

To display the plates and the responses, a C-language-based program was written to display the plates at pre-specified times and allow the subjects to select the desired option. Figure 1 illustrates a sample of the Ishihara test with possible options to answer. The specifications of the execution protocol, such as background color and sequence of plates displayed, were based on the protocol of the latest version of the Ishihara test. The program can display plates, record the subject's name, pixel resolution, and a timer to modify as well as to specify the display duration of each picture. For each picture, the possible options were written below the displayed picture. Among all four options, a Nothing option was inserted to capture the subjects' failure to detect any of the responses. To indicate the pictures at a higher resolution in this test, 800×800 pixels were selected.

Initially, each subject was placed in the resting position for 10 s. Then, they were asked to keep their eyes closed for another 10 s. There was a time delay of 3 s from clicking on the start option to display the first image in the C-based image player.

Each image was displayed to the subjects once for 12 s. Given the number of Ishihara test images and the time of the resting and delay mode, the total time was approximately 479 s ($10 \text{ s} + 10 \text{ s} + 3 \text{ s} + 38 \times 12 \text{ s}$). Figure 2 depicts a schematic of the task steps for each person.

2.2. Recording EEG signals

The brain signal was recorded by a NCC (Shanghai Ncc Electronic Corp) with 21 electrodes according to the international system of electrode placement (10–20). The 19 channels and two channels A1 and A2 as references are illustrated in figure 3. In this figure, in addition to showing the electrodes, the brain regions have also been classified according to [31]. The sampling frequency was 256 Hz. The mean values of

A1 and A2 electrodes, as the reference electrodes, were subtracted from the signal.

No electrical devices existed besides the subjects were considered, and the computer was on battery mode. Subjects were reminded to stay motionless and pay attention to the test. To ensure the optimal illumination, all experiments were conducted between 11:00 a.m. and 2:00 p.m, with the experiment performed in the room insulated from high-intensity lights and noise, where only moderate lights were used. Note that all procedures performed in studies involving the human members were in accordance with the ethical guidelines of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards [32]. all subjects signed informed consent for participation in the experiments.

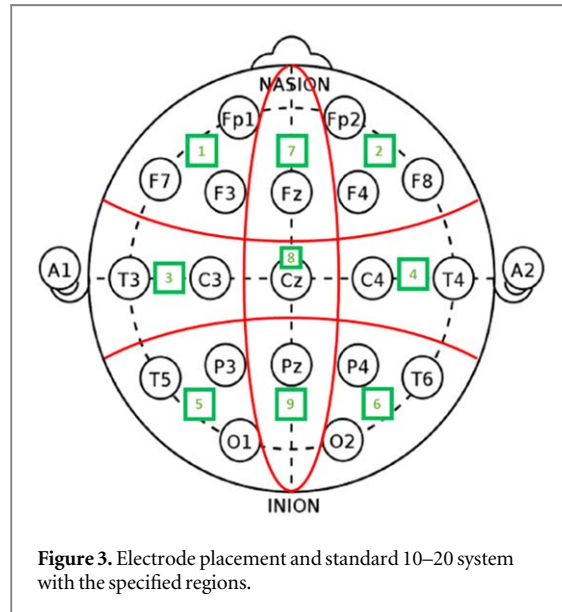
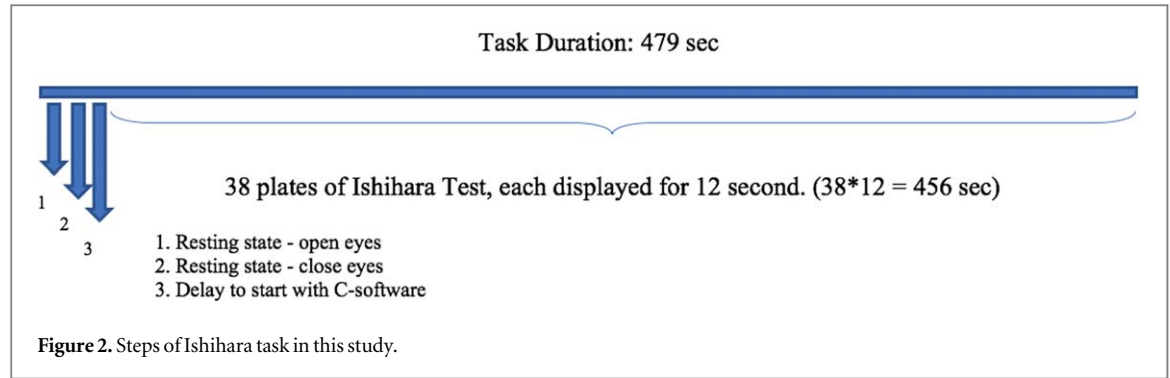
2.3. Signal Pre-processing and noise reduction

In the first step of pre-processing, a low-pass filter with a cut-off frequency of 40 Hz and a 0.1 dB ripple (Chebyshev) was used to clear the raw signal from the environmental noises, especially power line noise. A high-pass filter with a cutoff frequency of 1.5 Hz was selected to eliminate the noise caused by blinking and/or eye movements as well as head movements.

A C-based application was developed to show the images. All members of the healthy group answered the Ishihara test images correctly. This issue reaffirmed the vision of all members of the healthy group. On the other hand, all subjects in the target group responded correctly to 11 of the 38 images. It is meaning that the intensity of their color blindness did not affect recognizing the number or path in those images. The response of the target group was very variable in 14 images. Accordingly, between 1 to 9 people answered correctly in those 14 images and the rest answered wrongly. In other words, not everyone answered the same. Therefore, the remaining 13 images were images that all members of the target group made a mistake in responding to those images. The 38 images of the Ishihara test and the response of the control, as well as the target group to the images, are given in the Supplementary Material in figure S1 and tables S1–S2 (available online at stacks.iop.org/BPEX/7/025008/mmedia), respectively.

2.4. Feature extraction

The frequency features of the brain signals were used to investigate the differences between the two groups. Considering the limited frequency band of the brain signals, this signal can be subdivided into frequency bands. This division is generally based on the assumption that the brain signals become more active at some frequency intervals under certain conditions. There are different methods for estimation, modeling, and separation of frequency intervals of the brain signal. In this article, one of the most commonly used



parametric modeling methods, namely autoregressive (AR) modeling, was used to estimate the frequency spectrum.

$$X(t) = \sum_p^{k=1} A_k X(t - k) + e(t) \quad (1)$$

In this method, according to the formula, $X(t)$ is the brain signal at time t , and $e(t)$ represents the modeling error value, A_k denotes a matrix of coefficients multiplied by the past value of the signal considering the order of P Model. There are several methods such as the Akaike information criterion to consider the appropriate value for the model order. If the selected order is too high, the model becomes over fitted and represents noise, while if a low order is selected, the model will not be sufficient to represent the signal. By examining the Bode diagrams of EEG signals and observing the peaks in the power spectrum diagram in terms of frequency, the AR model with an order of 10 was chosen [33]. AR model estimation was used to separate the brain signals selected from each sample at delta (δ) frequency band (1.5–4 Hz), theta (θ) frequency band (4–8 Hz), alpha (α) frequency band (8–13 Hz), beta1 (β_1) frequency band (13–18 Hz), and beta2 (β_2) frequency band (18–30 Hz) [34, 35]. The mean absolute value of the frequency component

(frequency band) of each electrode was calculated and considered as the feature. This process was applied to brain responses to 13 pictures of 32 test pictures explained above.

2.5. Statistical test and classification

To compare the mean of the two samples apart from quantitative data, independent samples t-test is often used. The null hypothesis is the two groups are equal and there are no significant differences. The statistical test used in this study investigated the presence or absence of a significant difference ($P < 0.05$) between the target and control groups. The test was performed for all 19 electrodes in the frequency bands corresponding to the sample brain signals recorded. According to the statistical test results, the number of images that would make a significant difference between the two groups in each channel and frequency band was counted. It is worthwhile mentioning that because of the large number of statistical tests, the False Discovery Rate (FDR) correction is performed.

After identifying the electrodes and images that would show significant differences between the mean values of the features) average power frequency in each frequency band (in the two groups, a KNN classifier was used. The purpose of this step was to determine the best pictures in the Ishihara test, which can distinguish brain responses in the afferent electrodes in the two groups. For this purpose, KNN classifiers with K values of 3, 5, and 7 were used. Classification accuracy results were averaged for all of the three K values for each of the 13 images in the selected electrodes. Features (Δ , θ , β_1 , β_2) were applied as inputs to the KNN classifier (features that showing the highest difference in the statistical test). Finally, 12 features were employed for the classification process between the two groups using KNN. (three electrodes with each containing four features). Considering the limited number of subjects, the Leave One Out (LOO) method was used to train and test the KNN classifier.

3. Results

After constructing the feature vector for all subjects, statistical differences were compared between the two

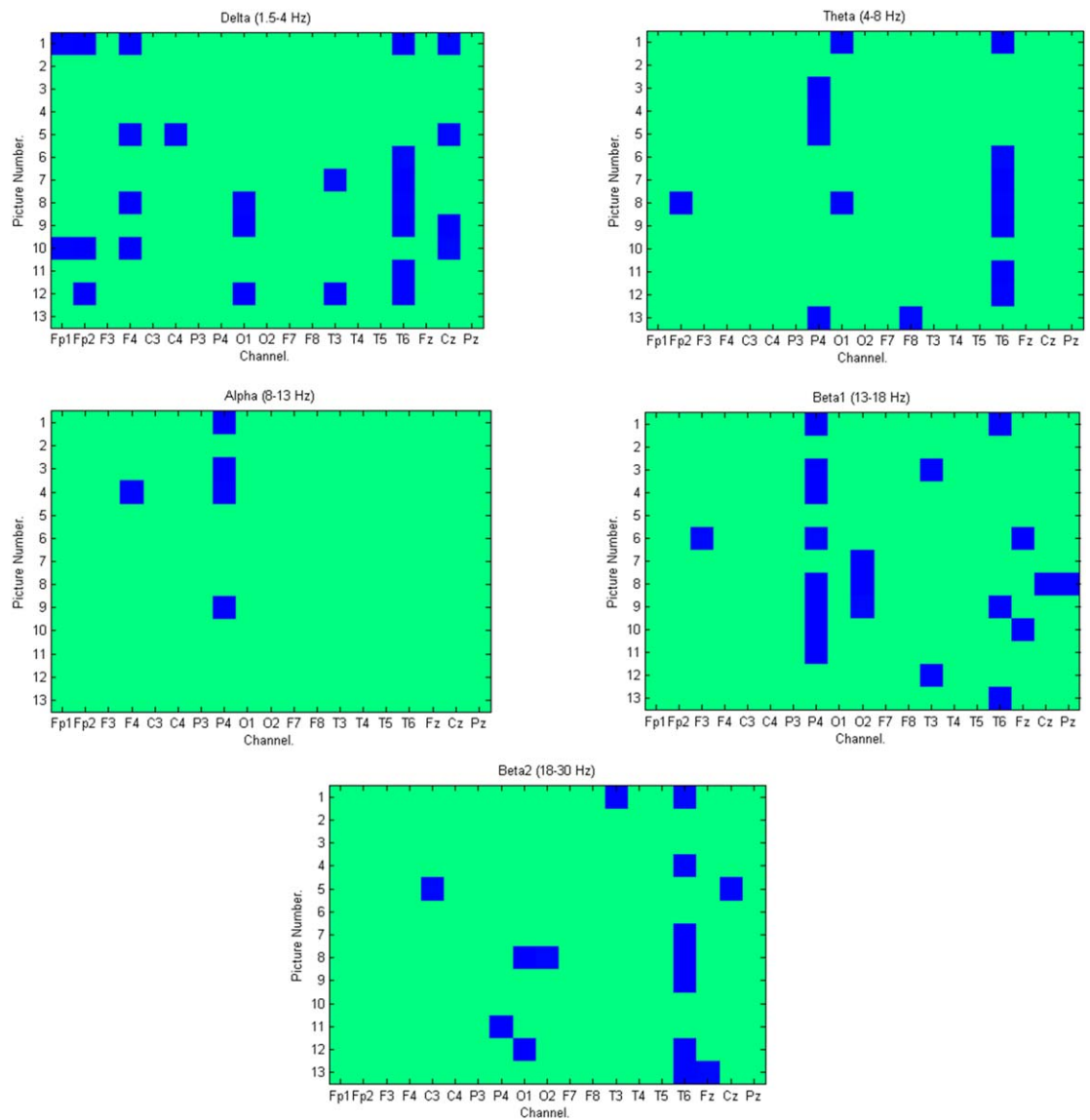


Figure 4. Statistical analysis of responses to pictures across all electrodes between the two groups (blue: significant difference, and green: no significant difference).

Table 2. Number of images that reporting statistically significant differences between the two groups (from 13 images selected) in each channel in each frequency band.

Channel	<i>Fp1</i>	<i>Fp2</i>	<i>F3</i>	<i>F4</i>	<i>C3</i>	<i>C4</i>	<i>P3</i>	<i>P4</i>	<i>O1</i>	<i>O2</i>
Frequency Band										
Delta	2	3	0	4	1	1	0	0	3	0
Theta	0	1	0	0	0	0	0	4	2	0
Alpha	0	0	0	1	0	0	0	4	0	0
Beta1	0	0	1	0	0	0	0	8	0	3
Beta 2	0	0	0	0	1	0	0	1	2	1
Channel	<i>F7</i>	<i>F8</i>	<i>T3</i>	<i>T4</i>	<i>T5</i>	<i>T6</i>	<i>Fz</i>	<i>Cz</i>	<i>Pz</i>	
Frequency Band										
Delta	0	0	2	0	0	7	0	4	0	
Theta	0	1	0	0	0	7	0	0	0	
Alpha	0	0	0	0	0	0	0	0	0	
Beta1	0	0	2	0	0	3	2	1	1	
Beta 2	0	0	1	0	0	7	1	1	0	

Table 3. Number of images that reporting statistically significant differences between the two groups (from 13 images selected) in each region in each frequency band

Region Frequency band	1	2	3	4	5	6	7	8	9	Total in each frequency band
Delta	2	7	3	1	3	7	0	4	0	22
Theta	0	2	0	0	2	11	0	0	0	15
Alpha	0	1	0	0	0	4	0	0	0	5
Beta1	1	0	2	0	0	14	2	1	1	20
Beta2	0	0	2	0	2	9	1	1	0	15
Total in each region	3	10	7	1	7	55	3	5	1	

Table 4. Accuracy of KNN classifier ($K = 5$) for the selected three electrodes (the numbers are the mean accuracy in percentage)

Picture	10	11	12	13	22	23	24	25	26	27	30	31	32
Mean classification accuracy	85.2	80.3	81.8	81	83.5	79.5	81.8	70.3	83.1	78.9	78.3	80.7	81.1
STD	2.2	2.6	3	3.1	2.7	3.3	2.9	5.4	2.5	2.8	2.9	3	2.8

groups. Figure 4 depicts the results of the statistical test. This figure contains 5 sections, each showing statistical test results in a specific frequency band. The horizontal axis in each one indicates the electrode number while the vertical axis represents the picture number. The blue sections indicate a significant difference between the target and control groups in terms of the mean feature value in a given electrode in response to a given picture.

Table 2 reports the three electrodes with the highest number of images causing a statistically significant difference. P4 Electrode at θ , α , and β_1 frequency bands reveal the largest difference in the statistical test between the two groups (figure 4). Also, T6 Electrode at δ , θ , and β_2 frequency bands also indicates a significant difference in the statistical test compared to other electrodes. O1 Electrode also showed the maximum difference at the δ frequency band, while representing such a significant difference in θ and β_2 frequency bands in response to fewer pictures.

The statistical test revealed a significant difference between the two groups, in P4, T6, and O1 electrodes over the entire four frequency bands (δ , θ , β_1 , and β_2). Considering the brain regions in figure 3, the number of channels that were statistically different in each region, and each frequency band have been counted in table 3.

In the classification stage, according to the results of t, the features of channels with more significant differences were selected (P4, T6, O1). The alpha band features for all three electrodes were removed and the four frequency band features (δ , θ , β_1 , and β_2) were used for classification.

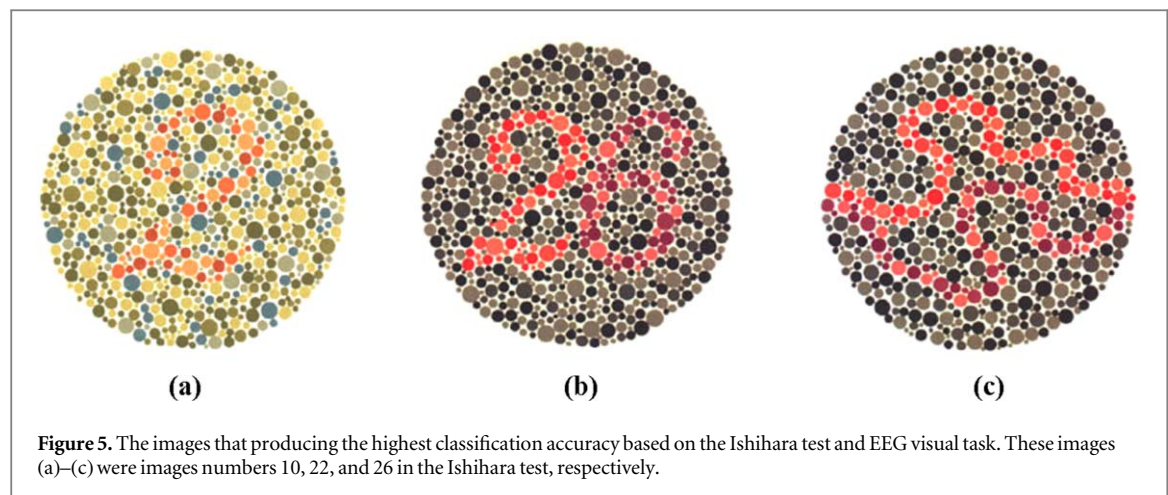
This process was performed for each of the 13 selected pictures to identify the picture yielding the most accurate classification results. According to the results, three pictures had the highest accuracy in classifying the two groups (table 4). The best result was to use the algorithm with a value of $K = 5$. Figure 5

illustrates the three pictures of the Ishihara test that presented the best classification accuracy.

Figure 5 displays that the classification accuracy of the two groups in the three images 10, 22, and 26 is higher than the other images. According to the results of answering the test in Table S2, provided in the supplementary material, all members of the target group could not recognize the number inside figure 5(a) and chose 'Nothing' for this image as well as for similar images (image 11–13 in figure S1). The results of table S2 showed that all subjects in the target group chose the wrong option in response to images figures 5(b) and (c) and did not recognize the purple line or number.

4. Discussion

In this study, to examine the differences in the EEG of subjects with CVD and healthy subjects, the EEG of the two groups was recorded during the Ishihara test. Then, the preprocessing and noise reduction steps were performed on the data. In the next step, the 4-second EEGs of all individuals were separated from 38 plates in response to 13 plates. These images were images where the color blindness caused all members of the target group to choose the wrong option in the test. By dividing the signals into the delta, theta, alpha, beta1, and beta2 frequency bands, the average power frequency was extracted as a feature for each individual in each frequency band. Statistical testing with FDR correction was performed on the values of the features of the two groups in response to 13 plates in 19 electrodes. Table 2 showed that the P4 channel in the theta, alpha, and beta1 frequency bands, exhibits more significant differences in more images. Statistical testing also showed significant differences in the frequency features of the T6 channel in the delta, theta, beta1, and beta2 bands compared to other channels (in response to more images). The O1 channel claimed the third rank, which had the most significant



difference in the images in the frequency bands of the delta, theta, and gamma. Since the P4 and T6 electrodes are located next to each other in the 10–20 system, it can be concluded that the regions related to these two electrodes (approximately the right temporoparietal region) would show different performance during the Ishihara test between the two groups. This result was also shown in table 3. When brain channels were categorized regionally (as in figure 3), the number of images for which the channels in each region made a significant difference between the two groups was counted. Table 4 showed that region 6, which includes P4, T6, and O2 electrodes, demonstrates more significant differences compared to other brain regions. This table also showed that the alpha band made the least significant difference in the electrodes compared to other frequency bands.

According to table 2, the features of the channels that produce the most significant differences (P4, T6, O1) were selected in four frequency bands (δ , θ , β_1 , and β_2) for the classification stage. At this stage, the KNN ($K = 5$) classifier and the LOO method were used for cross-validation of the results. The results showed that figure 5(a), presenting the number 10 of the Ishihara test, produced the highest classification accuracy ($85.2\% \pm 2.2$) between the two groups. The second and third images with the highest classification accuracy were figure 5(b) and figure 5(c), which were able to separate the two groups with 83.5% and 83.1% accuracy, respectively. In these two images and similar images (image 22–27 in figure S1) where a subject with deuteranopia disorder could only recognize a red number or line and was unable to recognize the purple line or number. The results of option selection in the target group in table S2 in supplementary material confirmed this color vision disorder.

To the best of our knowledge, there is no specific article on EEG analysis during the Ishihara test to compare the results. As such, it can be concluded that this research provides new insights into the difference in brain function between subjects with CVD and healthy subjects based on EEG signals. In this study, it

was found that the EEG signals of the right temporoparietal region (especially P4 and T6) had the most significant difference between the groups as compared to the rest of the brain in the healthy and CVD groups in the Ishihara vision test had the most significant difference between the groups. By extracting the frequency features from the three channels (P4, T6, and O1) making the most significant difference, the classification process was performed and the two groups were distinguished from each other by the KNN classifier with 85.2% accuracy. Also, three images of the Ishihara test were identified, in response to which the classification of the selected EEG signals in response to them reported the highest accuracy.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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